

High-frequency data and limit order books



Lecture 6 - March 17th, 2026

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CentraleSupélec

U. Paris-Saclay CentraleSupélec cursus Ingénieur 3A Mathématiques et Data Science
IP Paris ENSAE M2 Statistics, Finance and Actuarial Science
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Table of contents

Market impact

Optimal execution

- Optimal execution in a basic framework
- Optimal execution with cost uncertainty
- Optimal execution in a block-shaped LOB

Propagator models

- A propagator model for trades
- A propagator model for LOB events

Lab 5

Table of contents

Market impact

Optimal execution

- Optimal execution in a basic framework
- Optimal execution with cost uncertainty
- Optimal execution in a block-shaped LOB

Propagator models

- A propagator model for trades
- A propagator model for LOB events

Lab 5

Market impact

Market impact

Market impact can be described as

- (i) the effect of an order on the market,
- (ii) and the response of the market to a given order.

- ▶ First aspect of market impact is purely **mechanical**.
- ▶ Market impact of a given order is affected by existing **correlations in order flows**:
 - ▶ liquidity taking/momentum trading,
 - ▶ liquidity providing/mean-reversion,
 - ▶ **metaorders** (large investment decisions split into **multiple child orders** sent successively to the exchange).
- ▶ Market impact can be described in terms of **quantities** (function of the size of the order/metaorder) and in terms of **time** (instantaneous, **permanent**, **transient**, etc.).

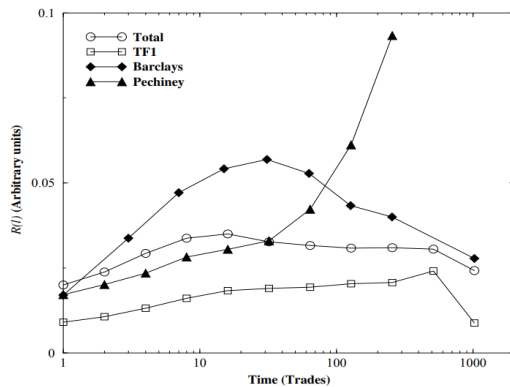
Response functions I

- Market impact can be analyzed via **response functions** [6]. If p_n is the mid-price just before the n -th trade et $\epsilon_n \in \{-1, 1\}$ the sign of the trade, the **response function of trades in trade time** is

$$\mathcal{R}(\ell) = \langle (p_{n+\ell} - p_n) \epsilon_n \rangle.$$

- This can be measured on the market but should not be interpreted as the impact of a single order as it is affected by all correlated subsequent orders.

Source: [6]



Response functions II

- ▶ The first value $\mathcal{R}(1)$ must be analyzed w.r.t. the **spread** and the **ticksize**.
- ▶ If V_n is the size of the n -th trade, a volume-dependent response function for the trades is:

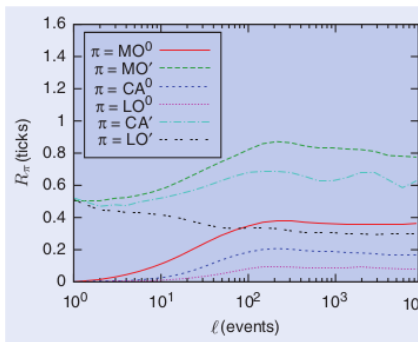
$$\mathcal{R}(\ell, V) = \langle (p_{n+\ell} - p_n) \epsilon_n | V_n = V \rangle.$$

Empirical studies on trades suggest that a variable separation is valid:

$$\mathcal{R}(\ell, V) = \mathcal{R}(\ell) f(V),$$

with $f(V)$ being strongly concave (suggested forms for $f(V)$: $V^{-\alpha}$ with $\alpha \approx 0.2 - 0.4$, $\log V$, etc.)

Response functions III



Source: [7]

- ▶ Response functions can be defined for any type of event affecting the market.
- ▶ The **response function** $R_\pi(\ell)$ describes the average price movement ℓ events after the occurrence of an order of a given type π :

$$R_\pi(\ell) = \langle (p_{t+\ell} - p_t) \epsilon_t \mid \pi_t = \pi \rangle$$

- ▶ *Left*: examples of response functions in event-time for LOB events.
- ▶ Again, this response function is a global market response that depends on the **correlated time structure of the order flows**.

Metaorders and the square-root law I

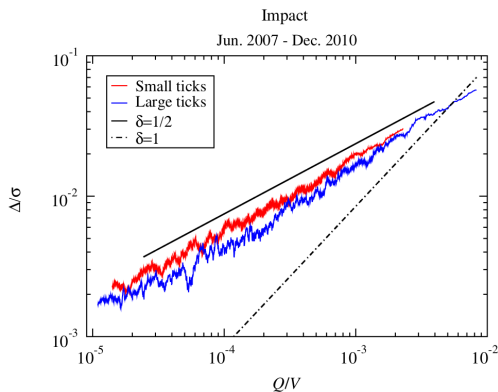
- ▶ **Metaorders** are large investment decisions split into **multiple child orders** sent successively to the exchange.
- ▶ Let $I(Q)$ be the average price variation between the first and last child orders of a metaorder of size Q . A widely-used transaction cost model assumes a **square-root law** for the price impact:

$$I(Q) = \alpha \sigma \sqrt{\frac{Q}{V}},$$

where α is a constant, σ is the **daily volatility** of the asset and V is the (average) **daily traded volume**.

- ▶ Many variants of the square-root law are implemented in trading firms/software.
- ▶ Puzzle: Rate of trading and optimal execution ?
- ▶ Empirical verification on proprietary datasets only.

Metaorders and the square-root law II



Source : [11]

- ▶ (Left) Empirical market impact of metaorders in CFM (2007-2010, proprietary data).
- ▶ Observations valid on equities, futures, cryptocurrencies. Universal ? [10]
- ▶ Modeling challenges: trading models compatible with a square-root (or power-law with an exponent smaller than one) impact ? arbitrage in these trading models ?
- ▶ Time-decay of the impact of metaorders ?
- ▶ Further reading: [5, 8, 4]

Table of contents

Market impact

Optimal execution

- Optimal execution in a basic framework

- Optimal execution with cost uncertainty

- Optimal execution in a block-shaped LOB

Propagator models

- A propagator model for trades

- A propagator model for LOB events

Lab 5

Table of contents

Market impact

Optimal execution

- Optimal execution in a basic framework

- Optimal execution with cost uncertainty

- Optimal execution in a block-shaped LOB

Propagator models

- A propagator model for trades

- A propagator model for LOB events

Lab 5

A basic framework with linear impact I

- ▶ Early contribution [3]
- ▶ A trader needs to buy a **total quantity** Q_T before a (finite) **time horizon** T and may trade during N trading periods $[t_n, t_{n+1})$, $n = 0, \dots, N-1$ with $t_n = n\frac{T}{N}$, $n = 0, \dots, N$.
- ▶ At t_n , $n = 0, \dots, N-1$, the trader decides the number of shares q_n to be bought (> 0) or sold (< 0) in the coming period $[t_n, t_{n+1})$.
- ▶ The set of admissible (deterministic) execution strategies is therefore

$$\Theta = \left\{ (q_0, \dots, q_{N-1}) \in \mathbb{R}^N : \sum_{n=0}^{N-1} q_n = Q_T \right\}.$$

- ▶ Let R_n be the trader's **remaining number of shares to purchase** at time t_n . We have

$$R_0 = Q_T, \quad R_n = R_{n-1} - q_{n-1}, \quad R_N = 0.$$

A basic framework with linear impact II

- ▶ Let P_n be the execution price for the period $[t_n, t_{n+1})$. The basic framework assumes a permanent linear impact with coefficient $\theta > 0$:

$$P_n = P_{n-1} + \theta q_n + \varepsilon_n,$$

where ε_n 's are i.i.d. with $\mathbf{E}[\varepsilon_n \mid q_n, P_{n-1}] = 0$. P_{-1} is the price before trading begins.

- ▶ Minimizing the trading costs means solving the problem

$$\min_{(q_0, \dots, q_{N-1}) \in \Theta} \mathbf{E} \left[\sum_{n=0}^{N-1} P_n q_n \right]$$

- ▶ This optimization problem can be solved with the dynamic programming principle.

A basic framework with linear impact III

- The optimal-value function satisfies the **Bellman equation**:

$$V_n(P_{n-1}, R_n) = \min_{q_n} \mathbf{E}_n[P_n q_n + V_{n+1}(P_n, R_{n+1})]$$

- **Terminal condition**: in the final trading period $[t_{N-1}, t_N)$, all the remaining shares must be bought, so that $q_{N-1} = R_{N-1}$ (no choice) and

$$V_{N-1}(P_{N-2}, R_{N-1}) = \mathbf{E}_{N-1}[P_{N-2} R_{N-1}] = (P_{N-2} + \theta R_{N-1}) R_{N-1}$$

- For the previous trading period, the Bellman equation is

$$V_{N-2}(P_{N-3}, R_{N-2}) = \min_{q_{N-2}} \mathbf{E}_{N-2}[P_{N-2} q_{N-2} + V_{N-1}(P_{N-2}, R_{N-2} - q_{N-2})]$$

A basic framework with linear impact IV

- By substituting V_{N-1} and P_{N-2} , we obtain a quadratic function of q_{N-2} that is easily minimized. We obtain the optimal values

$$\begin{cases} q_{N-2}^* = \frac{R_{N-2}}{2} \\ V_{N-2}^*(P_{N-3}, R_{N-2}) = R_{N-2}(P_{N-3} + \frac{3}{4}\theta R_{N-2}) \end{cases}$$

- Iterating backwards, the optimal values at step $N - k$ are

$$\begin{cases} q_{N-k}^* = \frac{R_{N-k}}{k} \\ V_{N-k}^*(P_{N-k-1}, R_{N-k}) = R_{N-k}(P_{N-k-1} + \frac{k+1}{2k}\theta R_{N-k}) \end{cases}$$

A basic framework with linear impact V

- ▶ Finally, the optimal strategy in this context is to evenly split trading quantities across trading periods:

$$q_n^* = \frac{R_0}{N} = \frac{Q_T}{N}, \quad n = 0, \dots, N-1,$$

and the trading cost for this associated to this optimal execution is

$$V_0^* = Q_T P_{-1} + \frac{\theta Q_T^2}{2} \left(1 + \frac{1}{N} \right)$$

- ▶ The greater the number of trading periods (horizon) available, the lower the trading cost, but the cost does not vanish completely (**permanent impact**)

Table of contents

Market impact

Optimal execution

Optimal execution in a basic framework

Optimal execution with cost uncertainty

Optimal execution in a block-shaped LOB

Propagator models

A propagator model for trades

A propagator model for LOB events

Lab 5

Optimal execution as a mean-variance problem I

- ▶ Framework proposed in [2] (known as the Almgren-Chriss model).
- ▶ At time 0 a trader holds an initial inventory of Q_0 shares and needs to liquidate it by a (finite) time horizon T .
- ▶ N trading periods of length $\tau = T/N$. Set $t_n = n\tau$, $n = 0, \dots, N$.
- ▶ At the start of the trading period $[t_n, t_{n+1})$, the trader decides the number of shares q_{n+1} to be bought (> 0) or sold (< 0) in the coming period.
- ▶ The trading rate is

$$\nu_{n+1} = \frac{q_{n+1}}{\tau}.$$

- ▶ The set of admissible (deterministic) execution strategies is therefore

$$\Theta = \left\{ (\nu_1, \dots, \nu_N) \in \mathbb{R}^N : \sum_{n=0}^{N-1} \nu_{n+1} \tau = -Q_0 \right\}.$$

Optimal execution as a mean-variance problem II

- Let Q_n be the trader's **inventory** at time t_n . We have

$$Q_{n+1} = Q_n + \nu_{n+1}\tau.$$

- In the absence of action by the trader, the stock (mid-)price would be a random walk (**fundamental value**). The model assumes a **permanent linear impact** of the trader's actions. The (mid-)price dynamics is thus

$$S_{n+1} = S_n + \sigma\sqrt{\tau}Z_{n+1} + \gamma\nu_{n+1}\tau,$$

for $\sigma > 0$, $\gamma > 0$ and $(Z_n)_n$ a sequence of i.i.d. random variables with standard Gaussian distribution.

Optimal execution as a mean-variance problem III

- There is no model of the limit order book. At each trading period, the trader will face execution costs (spread, walking the LOB and/or temporary impact). The **average execution price per share** on the trading period $[t_n, t_{n+1})$:

$$\tilde{S}_{n+1} = S_n + \eta \nu_{n+1}$$

for some $\eta > 0$.

- The dynamics of the **cash position** of the trader is therefore

$$X_{n+1} = X_n - \nu_{n+1} \tau \tilde{S}_{n+1} = X_n - \nu_{n+1} S_n \tau - \eta \nu_{n+1}^2 \tau.$$

- We assume that the trader seeks to maximize the expected final wealth while minimizing risk, i.e. has the objective function

$$\mathbf{E}[X_N] - \lambda \mathbf{V}[X_N].$$

for some **risk-aversion parameter** $\lambda > 0$.

Optimal execution as a mean-variance problem IV

- Direct computations for the terminal wealth of the trader yields:

$$X_N = X_0 + Q_0 S_0 - \frac{\gamma}{2} Q_0^2 + \sigma \sqrt{\tau} \sum_{i=0}^{N-1} Q_{n+1} Z_{n+1} - \sum_{i=0}^{N-1} \nu_{n+1}^2 \overbrace{\left(\eta - \frac{\gamma \tau}{2} \right)}^{\eta'} \tau.$$

- The trader thus faces the optimization problem

$$\min_{\nu \in \Theta} \left\{ \sum_{i=0}^{N-1} \nu_{n+1}^2 \eta' \tau + \lambda \sigma^2 \tau \sum_{i=0}^{N-1} Q_{n+1}^2 \right\},$$

which is rewritten w.r.t. the variables Q_n as:

$$\min_{Q \in \Theta_Q} \left\{ \frac{\eta'}{\tau} \sum_{i=0}^{N-1} (Q_{n+1} - Q_n)^2 + \lambda \sigma^2 \tau \sum_{i=0}^{N-1} Q_{n+1}^2 \right\}$$

Optimal execution as a mean-variance problem V

over the set of admissible inventory curves

$$\Theta_Q = \left\{ (Q_1, \dots, Q_{N-1}) \in \mathbb{R}^{N-1} \right\}.$$

(Q_0 given and $Q_N = 0$).

- If $\eta' = \eta - \frac{\gamma\tau}{2} > 0$, the objective is convex and the optimal solution can be explicitly computed

$$Q_n^* = Q_0 \frac{\sinh(\kappa(T - t_n))}{\sinh(\kappa T)}$$

where κ satisfies:

$$\frac{2}{\tau^2} (\cosh(\kappa\tau) - 1) = \frac{\lambda\sigma^2}{\eta'} = \frac{\lambda\sigma^2}{\eta - \frac{\gamma\tau}{2}}.$$

Optimal execution as a mean-variance problem VI

► Typical optimal trading curve in the Almgren-Chriss framework

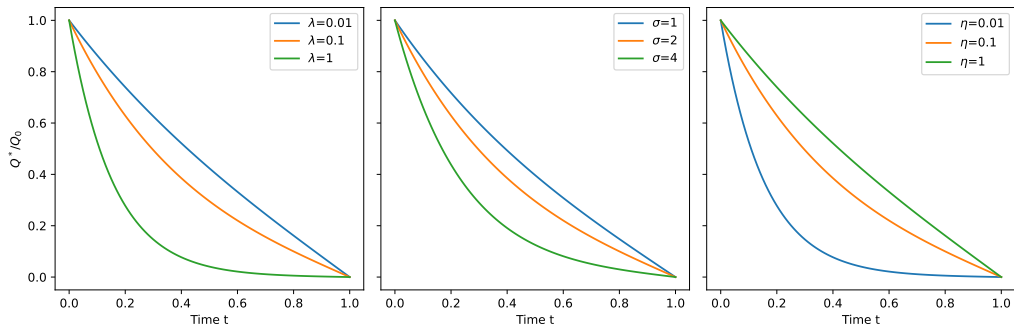


Table of contents

Market impact

Optimal execution

Optimal execution in a basic framework

Optimal execution with cost uncertainty

Optimal execution in a block-shaped LOB

Propagator models

A propagator model for trades

A propagator model for LOB events

Lab 5

Optimal execution in a block-shaped LOB I

- ▶ Proposed by Obizhaeva and Wang in [9] (originally released as a preprint in 2005), extended in [1] to more general shapes.
- ▶ A trader needs to buy a **total quantity** X_0 before a (finite) **time horizon** T and may trade at discrete times $0 = t_0 < t_1 < \dots < t_{N-1} < t_N = T$.
- ▶ Set of possible strategies: $\Theta = \left\{ x = (x_{t_0}, \dots, x_{t_N}) \in [0, +\infty)^{N+1} : \sum_{i=0}^N x_{t_i} = X_0 \right\}$.
- ▶ The model assumes **continuous prices** (a_t ask price at time t) and a **block-shaped LOB with available size q at all prices**. On the ask side, the LOB is represented at time t by

$$q_t(p) = q \mathbf{1}_{[a_t, +\infty)}(p)$$

- ▶ There exists a "**fundamental**" **mid-price** f_t (a Brownian motion). *In the absence of trades*, the LOB moves with f_t at all times with a constant spread s .

Optimal execution in a block-shaped LOB II

- ▶ At time $t_0 = 0$, LOB is defined with ask price $a_{t_0} = f_{t_0} + \frac{s}{2}$ and ask LOB density $q_{t_0}(p) = q \mathbf{1}_{[a_{t_0}, +\infty)}(p)$.
- ▶ If the trader trades the quantity x_{t_0} at $t_0 = 0$, then the new ask price satisfies

$$x_{t_0} = \int_{a_{t_0}}^{a_{t_0}+} q_{t_0}(p) dp, \quad \text{i.e.} \quad a_{t_0+} = f_0 + \frac{s}{2} + \frac{x_{t_0}}{q},$$

and the average execution price $\overline{p}_{t_0}$ of this trade satisfies

$$\overline{p}_{t_0} x_{t_0} = \int_{a_{t_0}}^{a_{t_0}+} p q_{t_0}(p) dp, \quad \text{i.e.} \quad \overline{p}_{t_0} = f_{t_0} + \frac{s}{2} + \frac{x_{t_0}}{2q}.$$

Optimal execution in a block-shaped LOB III

- ▶ The model assumes a **permanent linear impact** on the mid-price : *in the absence of other trades*, as $t \rightarrow +\infty$, the LOB would be defined with

$$a_t - f_t \xrightarrow{t \rightarrow +\infty} \frac{s}{2} + \lambda x_{t_0} \quad \text{and} \quad q_t(p) = q \mathbf{1}_{[a_t, +\infty)},$$

for a given $\lambda \in \left[0, \frac{1}{q}\right]$.

- ▶ The LOB is assumed to be **resilient** and liquidity returns **exponentially fast**: *in the absence of other trades*,

$$a_t = f_t + \lambda x_{t_0} + \frac{s}{2} + x_{t_0} \overbrace{\left(\frac{1}{q} - \lambda\right)}^{\kappa :=} e^{-\rho t},$$

for some $\rho > 0$.

Optimal execution in a block-shaped LOB IV

- Now, considering all trades on $[0, t)$, the ask price at time t becomes

$$a_t = f_t + \sum_{t_i < t} \lambda x_{t_i} + \frac{s}{2} + \sum_{t_i < t} x_{t_i} \kappa e^{-\rho(t-t_i)}.$$

- The ask price after trading x_{t_n} at time t_n is $a_{t_n+} = a_{t_n} + \frac{x_{t_n}}{q}$ and the average cost of this trade is $\overline{p}_{t_n} = a_{t_n} + \frac{x_{t_n}}{2q}$. Hence, the optimal trading problem is written

$$\min_{x \in \Theta} \mathbf{E} \left[\sum_{i=0}^N x_{t_n} \left(f_{t_n} + \sum_{i=0}^{n-1} \lambda x_{t_i} + \frac{s}{2} + \sum_{i=0}^{n-1} x_{t_i} \kappa e^{-\rho(t_n-t_i)} + \frac{x_{t_n}}{2q} \right) \right].$$

Optimal execution in a block-shaped LOB V

- This problem can be explicitly solved.

Sketch of the proof: using the new variables $X_{t_n} = X_0 - \sum_{i=0}^{n-1} \lambda x_{t_i}$, (remaining quantity to be traded remaining before time t_n) and $D_{t_n} = \sum_{i=0}^{n-1} x_{t_i} \kappa e^{-\rho(t-t_i)}$, ("deviation" from the impacted steady state), the problem takes a simpler form:

$$\min_{x \in \Theta} \mathbf{E} \left[\sum_{i=0}^N x_{t_n} \left(f_{t_n} + \lambda(X_0 - X_{t_n}) + \frac{s}{2} + D_{t_n} + \frac{x_{t_n}}{2q} \right) \right].$$

This can be solved using a backward induction (dynamic programming principle). One finds that the optimal traded quantity x_{t_n} at time t_n is linear in X_{t_n} and D_{t_n} . For explicit formulas, see [9, Proposition 1].

Optimal execution in a block-shaped LOB VI

- Typical shape of the optimal trading strategy in the Obizhaeva & Wang framework:

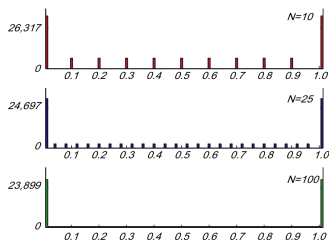


Fig. 2. Optimal strategy with N fixed discrete trading intervals. This figure plots the optimal trades for N fixed intervals, where N is 10, 25, and 100 for respectively the top, middle and bottom panels. The initial order to trade is set at $X_0 = 100,000$ units, the time horizon is set at $T = 1$ day, the market depth is set at $q = 5,000$ units, the price-impact coefficient is set at $\lambda = 1/(2q) = 10^{-4}$, and the resilience coefficient is set at $\rho = 2.231$.

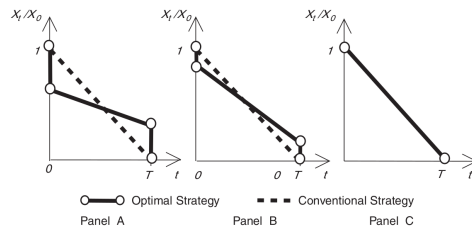


Fig. 4. Optimal strategy versus simple strategy from the conventional models. The figure plots the time paths of remaining order to be executed for the optimal strategy (solid line) and the simple strategy obtained from the conventional models (dashed line), respectively. The order size is set at $X_0 = 100,000$, the initial ask price is set at \$100, the market depth is set at $q = 5,000$ units, the (permanent) price-impact coefficient is set at $\lambda = 1/(2q) = 10^{-4}$, and the trading horizon is set at $T = 1$ day, which is assumed to be 6.5 hours (390 minutes). Panels A, B, and C plot the strategies for $\rho = 0.001, 2$ and $1,000$, respectively.

Source: [9]

Table of contents

Market impact

Optimal execution

Optimal execution in a basic framework

Optimal execution with cost uncertainty

Optimal execution in a block-shaped LOB

Propagator models

A propagator model for trades

A propagator model for LOB events

Lab 5

Table of contents

Market impact

Optimal execution

Optimal execution in a basic framework

Optimal execution with cost uncertainty

Optimal execution in a block-shaped LOB

Propagator models

A propagator model for trades

A propagator model for LOB events

Lab 5

A propagator models for trades I

- ▶ Initially introduced in [6].
- ▶ A **propagator model for trades** define the dynamics of the mid-price **in trade time** as a sum of the impacts of past trades and some noise:

$$p_n = p_{-\infty} + \sum_{m < n} [g(n - m)\epsilon_m + \eta_m],$$

where p_n is the mid-price just before the n -th trade, ϵ_n is the sign of the n -th trade, g is the **propagator** function (isolated impact of a single trade as a function of time) and η_m 's are exogenous i.i.d. noise terms.

A propagator models for trades II

- ▶ The response function $\mathcal{R}(\ell) = \mathbf{E}[(p_{n+\ell} - p_n)\epsilon_n]$ becomes in this model

$$\mathcal{R}(\ell) = g(\ell) + \sum_{0 < k < \ell} g(\ell - k)C(k) + \sum_{k > 0} [g(k + \ell) - g(k)]C(k),$$

where C defined by $C(m - n) = \mathbf{E}[\epsilon_m \epsilon_n]$ is the autocorrelation of trade signs.

- ▶ If ϵ_s 's are i.i.d, the $\mathcal{R}(\ell) = g(\ell)$. If furthermore the impact is permanent, the response function is constant.
- ▶ Similarly, setting $h(\ell) = g(\ell + 1) - g(\ell)$, the lag-1 return r_n is

$$r_n = p_{n+1} - p_n = g(1)\epsilon_n + \sum_{k < n} h(n - k)\epsilon(k) + \eta_n,$$

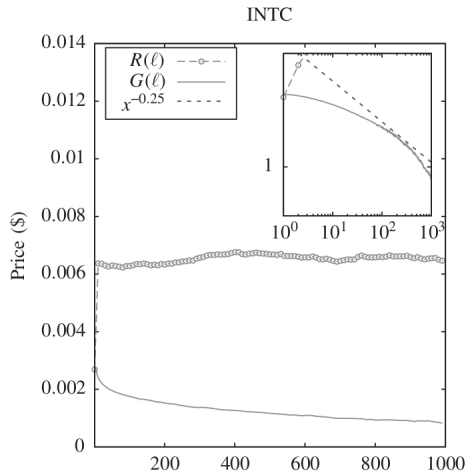
- ▶ Difficulty: ϵ_n 's are correlated while r_n 's should not be.

A propagator models for trades III

- ▶ This propagator model for trades can be calibrated using the sign-return lagged covariance:

$$\begin{aligned} C^{r,\epsilon}(\ell) &= \mathbf{E}[r_{n+\ell} \epsilon_n] \\ &= \sum_{m \geq 0} h(m) C(|m - \ell|). \end{aligned}$$

Numerically, by truncation and matrix inversion, we obtain h and then g .



Source: [4].

Table of contents

Market impact

Optimal execution

- Optimal execution in a basic framework

- Optimal execution with cost uncertainty

- Optimal execution in a block-shaped LOB

Propagator models

- A propagator model for trades

- A propagator model for LOB events

Lab 5

A propagator model with for LOB events I

- ▶ [7] proposes several **propagator models** to analyze impact of LOB events.
- ▶ Orders are classified by **type** (limit, market, cancel) and **aggressiveness** (the order moves the price, or does not move the price).
- ▶ Any order occurring at time t (*here in event time*) has a type π_t , a sign ϵ_t , a side s_t and a gap Δ_t :

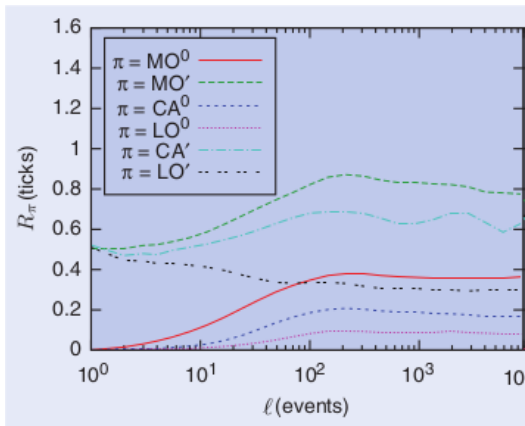
π	Event definition	Event sign definition	Gap definition ($\Delta_{\pi,\epsilon}$)
$\pi = \text{MO}^0$	Market order, volume < outstanding volume at the best	$\epsilon = \pm 1$ for buy/sell market orders	0
$\pi = \text{MO}'$	Market order, volume $\geq$ outstanding volume at the best	$\epsilon = \pm 1$ for buy/sell market orders	Half of the first gap behind the ask ($\epsilon = 1$) or bid ($\epsilon = -1$)
$\pi = \text{CA}^0$	Partial cancellation of the bid/ask queue	$\epsilon = \mp 1$ for buy/sell side cancellation	0
$\pi = \text{LO}^0$	Limit order at the current best bid/ask	$\epsilon = \pm 1$ for buy/sell limit orders	0
$\pi = \text{CA}'$	Complete cancellation of the best bid/ask	$\epsilon = \mp 1$ for buy/sell side cancellation	Half of the first gap behind the ask ($\epsilon = 1$) or bid ($\epsilon = -1$)
$\pi = \text{LO}'$	Limit order inside the spread	$\epsilon = \pm 1$ for buy/sell limit orders	Half distance of the limit order from the earlier best quote on the same side

A propagator model with for LOB events II

- ▶ The **response function** $\mathcal{R}_\pi(\ell)$ describes the average price movement ℓ events after the occurrence of an order of a given type π :

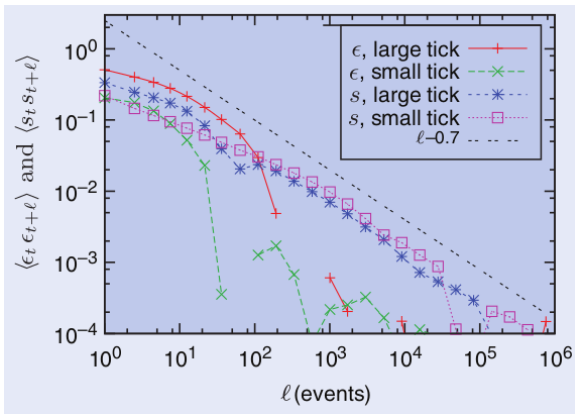
$$\mathcal{R}_\pi(\ell) = \langle (p_{t+\ell} - p_t)\epsilon_t \mid \pi_t = \pi \rangle$$

- ▶ This response function is a global market response : it is the combination of a “bare” impact of the order and the correlation functions.



Source: [7]

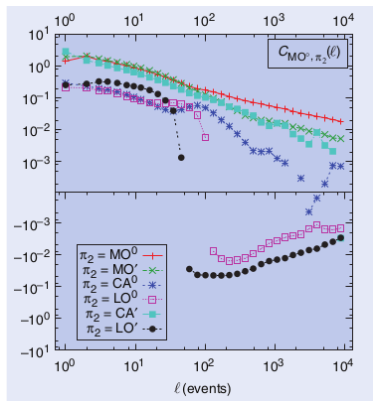
A propagator model with for LOB events III



Source: [7]

- Side autocorrelations exhibit long range memory: market orders persistently hit one side, and attract liquidity on the same side.

A propagator model with for LOB events IV



- A signed event-event correlation function is defined as:

$$C_{\pi_1, \pi_2}(\ell) = \frac{\langle \mathbf{1}_{\pi_t = \pi_1} \epsilon_t \mathbf{1}_{\pi_{t+\ell} = \pi_2} \epsilon_{t+\ell} \rangle}{\mathbf{P}(\pi_1) \mathbf{P}(\pi_2)}$$

Source: [7]

A propagator model with for LOB events V

- An exact formula for the mid-price movement simply tracks the price changes:

$$p_{t+\ell} = p_t + \sum_{t \leq t' < t+\ell} \epsilon_{t'} \Delta_{\pi_{t'}, \epsilon_{t'}, t'}.$$

- If we approximate mid-price movements per type of events by setting $\Delta_{\pi}^R = \langle \Delta_{\pi_{t'}, \epsilon_{t'}, t'} \mid \pi_{t'} = \pi \rangle$, we obtain a constant impact model for LOB events:

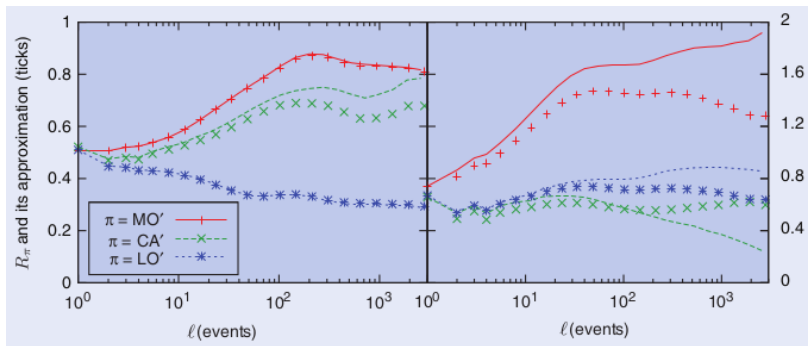
$$p_{t+\ell} \approx p_t + \sum_{t \leq t' < t+\ell} \epsilon_{t'} \Delta_{\pi}^R.$$

- The response function becomes:

$$\mathcal{R}_{\pi}(\ell) = \langle (p_{t+\ell} - p_t) \epsilon_t \mid \pi_t = \pi \rangle \approx \sum_{0 \leq t' < t+\ell} \sum_{\pi_1} \Delta_{\pi_1}^R \mathbf{P}(\pi_1) C_{\pi, \pi_1}(t').$$

A propagator model with for LOB events VI

- The model works very well for large tick stocks (dense LOB), but less for small tick stocks (rough gap approximation).



Source: [7]

Table of contents

Market impact

Optimal execution

- Optimal execution in a basic framework

- Optimal execution with cost uncertainty

- Optimal execution in a block-shaped LOB

Propagator models

- A propagator model for trades

- A propagator model for LOB events

Lab 5

Lab 5 - Questions

1. **A propagator model for trades.** For each stock of your dataset,
 - 1.1 compute the response function for trades in trade time ;
 - 1.2 calibrate a propagator model for trades ;
 - 1.3 comment.
2. **A propagator model for LOB events.** Select one stock of your dataset (stock0 or stock1).
 - 2.1 Plot the sign $\langle \epsilon_t, \epsilon_{t+\ell} \rangle$ and side autocorrelation $\langle s_t, s_{t+\ell} \rangle$ functions.
 - 2.2 Compute and plot the empirical response functions $\mathcal{R}_\pi(\ell)$ for each of the 6 types of events π . Comment.
 - 2.3 Compute the signed event-event correlations $C_{\pi_1, \pi_2}(\ell)$.
 - 2.4 Plot the function $C_{MO', \pi_2}(\ell)$. Note that a signed log-scale similar to the one presented in [7] can be obtained in matplotlib with `ax.set_yscale('symlog')`.
 - 2.5 Compute the theoretical responses in the constant impact model, and compare them to the empirical ones. Comment.

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